
Newton School of Technology

Data Visualization & Analytics | Capstone 2 | Final Project Report

F1 Constructor Sponsorship & Investment ROI Analytics

*Post-2014 Hybrid Era | Growth, Consistency & Risk Intelligence for
Corporate Sponsors*

Sector Sports Analytics & Corporate Sponsorship Intelligence

Team ID Section C — Group G13

Institute Newton School of Technology

Faculty Mentor Archit Raj

Submission Date April 29, 2026

Team Members

Roles, responsibilities, and project ownership summary

Name	Role	Primary Responsibility
Priyanshu Tomar	Team Lead / ETL Lead	Dataset ownership, ETL pipeline, report co-authoring, PPT support
Piyush	Strategy Lead	Recommendation framing, business strategy, PPT co-ownership
Kushagra Maheshwari	EDA & Visualisation Lead	EDA support, dataset support, visual analysis, and dashboard coordination
Thejas	Analytics & Stats Lead	Statistical analysis, validation, and analytical interpretation
Anurag Kumar	Data Lead / Report Lead	Dataset sourcing, profiling, and report co-ownership with team lead
Aditya Srivastava	Tableau & PPT Lead	Tableau ownership, presentation design, and PPT finalisation

GitHub Repository https://github.com/r0c0y/SectionC_G13_F1

Tableau Dashboard <https://public.tableau.com/app/profile/kushagra.maheshwari4048/vizzes>

Dataset Source <https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020>

Contents

1	Executive Summary	5
1.1	The Business Challenge	5
1.2	Our Approach	5
1.3	Key Findings	5
1.4	Key Recommendations	6
2	Sector & Business Context	6
2.1	The Formula 1 Industry	6
2.2	The Decision-Maker	6
2.3	Why This Problem Was Chosen	7
2.4	Business Value of Solving This Problem	7
3	Problem Statement & Objectives	7
3.1	Formal Problem Definition	7
3.2	Project Scope	7
3.3	Success Criteria	8
4	Data Description	8
4.1	Dataset Source	8
4.2	Primary Tables Used	9
4.2.1	results.csv — Race Results (Core Table)	9
4.2.2	Other Tables Used	9
4.3	Data Limitations and Known Biases	9
5	Data Cleaning & ETL Pipeline	10
5.1	Pipeline Architecture Overview	10
5.2	Phase 1: Extraction (01_extraction.ipynb)	12
5.2.1	Why the 2014 Filter?	12
5.2.2	Handling \N Null Values	12
5.2.3	The Six-Table Join Strategy	12
5.3	Phase 2: Cleaning (02_cleaning.ipynb)	13
5.3.1	Pre-Cleaning Null Audit	13
5.3.2	Step 1: Column Pruning	14
5.3.3	Step 2: Data Type Standardisation	14
5.3.4	Step 3: Feature Engineering	14
5.3.5	Step 4: Pit Stop Data Cleaning	15
5.3.6	Cleaning Log Summary	15
6	KPI & Metric Framework	16
6.1	Why These KPIs?	16
6.2	KPI Definitions	16
6.3	Composite Investment Score	17
6.4	KPI Mapping to Project Objectives	18
7	Exploratory Data Analysis (EDA)	18
7.1	Trend Analysis: Points Concentration Over Time	18
7.2	Comparison Analysis: Constructor Performance Landscape	19

7.3	Distribution Analysis: Result Volatility	20
7.4	Correlation Analysis	20
7.5	EDA Summary: Five Key Findings	21
8	Statistical Analysis	21
8.1	Analysis 1: Volatility–Points Linear Regression	21
8.2	Analysis 2: Pit Stop Speed vs Points (Independent Samples t-Test) . .	22
8.3	Analysis 3: Multiple Regression — Drivers of Season Points	22
8.4	Analysis 4: Momentum Trend Analysis (Per-Team Linear Regression) .	23
8.5	Statistical Summary Table	24
9	Tableau Dashboard Design	24
9.1	Dashboard Objective	24
9.2	Dashboard 1: F1 Constructor Investment Scorecard	25
9.2.1	KPI Cards (Top Banner)	25
9.2.2	Chart 1: Growth ROI vs Consistency Scatter Plot	25
9.2.3	Chart 2: Investment Score Ranking Bar Chart	26
9.3	Dashboard 2: Risk-Adjusted Investment Portfolio Matrix	27
9.3.1	KPI Cards	27
9.3.2	Chart 1: Risk vs Return Scatter (Efficient Frontier)	27
9.3.3	Chart 2: Momentum Trend Bar	28
9.3.4	Chart 3: Volatility Score Stacked Bar	28
9.4	Dashboard 3: Operational Excellence & Race Craft Analytics	28
9.4.1	KPI Cards	29
9.4.2	Chart 1: Pit Efficiency vs Consistency Scatter	29
9.4.3	Chart 2: DNF Reliability Trend Waterfall	29
9.4.4	Chart 3: Pit Strategy by Season Pie Chart	29
9.4.5	Chart 4: Reliability Risk by Team Bar	29
9.5	Interactive Filters	29
10	Insights Summary	30
11	Recommendations	32
12	Impact Estimation	32
12.1	R1: McLaren Premium Partnership	33
12.2	R2: Diversified Portfolio	33
12.3	R3: Red Bull Renegotiation	33
13	Limitations	33
14	Future Scope	34
15	Conclusion	34
	Appendix	35
A	Data Dictionary (Full Column Definitions)	35
B	Tableau-Ready Output Files	36

C Python Library Stack	36
D Points Concentration Data Table (EDA Output)	37
Contribution Matrix	37

1 Executive Summary

For the Executive Reader

This report is written for a corporate CMO, investment manager, or brand strategist deciding where to allocate sponsorship budgets in Formula 1. No prior knowledge of motorsport or data science is required.

In one line: this report helps sponsors identify which F1 teams offer the best mix of growth, stability, and value before committing large partnership budgets.

1.1 The Business Challenge

Formula 1 is a \$2.6 billion annual sponsorship market. For brands, an F1 deal can deliver global TV visibility, association with engineering excellence, and access to affluent international audiences.

The problem is that not every team offers the same value. Some teams are rising, some are plateauing, and some are declining. The difference between backing the right constructor and backing the wrong one can translate into tens of millions of dollars in lost exposure and weaker brand impact.

This project addresses one central gap: **sponsors lack a structured, data-driven way to compare teams on expected sponsorship return.** In practice, many decisions are still driven by reputation (“Ferrari is famous”) or recent headlines (“Red Bull just won everything”) instead of forward-looking performance evidence.

1.2 Our Approach

Using 11 seasons of race-by-race data from the Ergast F1 historical database (2014–2024), the team built a Python data pipeline and a three-dashboard Tableau suite. The workflow cleaned and analysed more than 4,600 race records covering 20 constructor teams, 59 drivers, and 32 circuits.

From that dataset, we measured five performance dimensions — growth, consistency, reliability, race craft, and operational excellence — and combined them into one composite **Investment Scorecard**. This gives sponsors a simpler way to compare teams using one clear, repeatable framework.

1.3 Key Findings

1. **McLaren is the strongest current sponsorship opportunity.** It is the only “Gold” tier team, with an Investment Score of 63.02, year-over-year points growth of +128.95%, and the steepest positive momentum slope in the dataset (+229.5 points per year). In plain terms, McLaren combines current performance with future upside.
2. **Red Bull looks expensive relative to its recent trajectory.** Although it dominated the 2022–2023 period, its momentum slope has turned sharply negative (−93.5 points per year) and its championship points fell by 32% in 2024. This suggests sponsors may be paying for brand prestige built on a cycle that is already

weakening.

3. **The grid is becoming more competitive.** The top team's share of total championship points fell from 36.6% in 2015 to 25.0% in 2024, the lowest level of the Hybrid Era. For sponsors, this means strong exposure is no longer limited to one dominant team; well-chosen mid-field partnerships now offer better value than before.
4. **Operational discipline is strongly linked to investment quality.** Lower result volatility ($R^2 = 0.28$, $p < 0.001$) and faster pit stops (Cohen's $d = 1.26$, $p < 0.001$) both predict stronger championship performance. Teams that execute cleanly and consistently are more likely to deliver dependable sponsor visibility.

1.4 Key Recommendations

1. **Prioritise McLaren for premium partnerships** — it currently offers the clearest blend of growth, consistency, and upside.
2. **Build a diversified F1 portfolio** (for example, McLaren + Sauber + Aston Martin) — this broadens exposure while reducing dependence on one team's results.
3. **Limit over-concentration in declining teams** such as Red Bull and Alpine until their momentum clearly improves over at least two consecutive seasons.

2 Sector & Business Context

2.1 The Formula 1 Industry

Formula 1 is the world's leading single-seater motorsport series, sanctioned by the FIA and operated commercially by Formula One Management (FOM). It runs roughly 24 races per season across six continents and delivers worldwide live broadcast exposure, which is why it remains one of the most valuable sponsorship platforms in sport.

The **Modern Hybrid Era**, which began in 2014, made the sport even more relevant to technology, engineering, and innovation-focused brands. Today's cars combine combustion power with advanced Energy Recovery Systems (ERS), so teams increasingly resemble high-performance technology businesses rather than traditional racing outfits.

2.2 The Decision-Maker

This project serves two primary stakeholder groups:

- **Corporate Chief Marketing Officers (CMOs)** allocating multi-million-dollar annual sponsorship budgets across motorsport properties.
- **Enterprise Investment Managers** managing sports investment portfolios and seeking to quantify the risk-adjusted return on F1 brand partnerships.

Both groups face the same issue: F1 sponsorship decisions are often driven by brand perception, relationships, and headline performance rather than a consistent evidence-based framework. There is no widely used public model that converts race data into clear investment intelligence.

2.3 Why This Problem Was Chosen

The team chose this problem because it combines *sports analytics*, *financial thinking*, and *brand strategy* in a way that is rarely done for Formula 1. The Ergast dataset provides rich race-by-race records, allowing us to calculate credible performance trends, volatility measures, and growth signals that are useful to real decision-makers.

2.4 Business Value of Solving This Problem

- **Direct cost savings:** A sponsor redirecting budget from a declining team (e.g., Red Bull in 2024) to a high-growth team (McLaren) could achieve an estimated 2–3× greater TV coverage per dollar spent, based on the ratio of race wins and podiums between the two teams.
- **Risk reduction:** Sponsors who use our Reliability Score and DNF analysis can avoid teams that, on average, fail to finish 14+ races per season — meaning they would have no TV coverage in those events.
- **Market timing advantage:** Identifying “undervalued gems” (teams with improving performance but below-average sponsorship premiums) before their growth becomes widely recognised allows sponsors to lock in long-term contracts at below-market rates.

For a sponsor, the business value is simple: choose teams early, choose them with evidence, and avoid paying premium rates for performance that may already have peaked.

3 Problem Statement & Objectives

3.1 Formal Problem Definition

Problem Statement: Corporate sponsors and investment managers allocating budgets to Formula 1 constructor partnerships currently lack a quantitative, data-driven framework to evaluate which teams deliver the highest risk-adjusted return on sponsorship investment. Decisions are made on brand perception rather than performance evidence, leading to misallocation of significant marketing budgets.

Core Business Question: “Which F1 constructor teams represent the highest Growth ROI and Consistency for corporate sponsorship in the post-2014 Modern Hybrid Era?”

3.2 Project Scope

In Scope:

- Race-by-race performance analysis for all 20 constructors competing in the 2014–2024 seasons.
- Constructor-level KPIs: points momentum, result volatility, finish rate, position delta, and pit stop efficiency.
- A composite Investment Scorecard ranking all teams on five weighted dimensions.
- Three interactive Tableau dashboards for executive and operational decision

support.

Out of Scope:

- Individual driver performance analysis (focus is constructor/team level).
- Financial modelling of actual sponsorship deal values (proprietary commercial data unavailable).
- Pre-2014 data (technically incomparable to modern hybrid-era performance metrics).
- Qualifying, practice, or sprint race data (focus is race-day performance).

3.3 Success Criteria

The project is considered successful if it produces:

1. A clean, reproducible Python ETL pipeline with documented transformation steps.
2. At least five statistically validated KPIs that directly map to investment decision-making.
3. A Tableau dashboard suite (three dashboards) that answers the business question without requiring the user to understand the underlying data.
4. A ranked list of constructor investment opportunities with actionable business recommendations.

4 Data Description

4.1 Dataset Source

Table 1: Dataset Overview

Attribute	Detail
Source Name	Ergast Developer API — Formula 1 Historical Database
Kaggle Link	https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020
Raw Files	14 CSV files
Raw Total Rows	701,433 (summation across all 14 files)
Raw Total Columns	120 (summation across all 14 files)
Analysis Period	2014–2024 (11 seasons, Modern Hybrid Era)
Filtered Rows	≈440,000 rows after era filter
Master Table (Post-Join)	4,626 rows × 35 columns
Active Constructors	20 unique teams
Active Drivers	59 unique drivers
Unique Circuits	32 circuits across 35 countries
Unique Races	228 race events

4.2 Primary Tables Used

Seven of the fourteen raw CSV files were selected as the analytical foundation, with **results.csv** serving as the core table. The remaining files act as lookup and enrichment tables joined on foreign keys.

4.2.1 results.csv — Race Results (Core Table)

One row per driver per race. This is the foundation of every KPI calculation.

Table 2: Key columns from results.csv

Column	Type	Description	Null %
resultId	int	Unique result identifier (PK)	0%
raceId	int	FK to races table	0%
driverId	int	FK to drivers table	0%
constructorId	int	FK to constructors; main grouping key	0%
grid	int	Starting grid position	0%
positionOrder	int	Final finishing order	0%
points	float	Championship points scored	0%
laps	int	Number of laps completed	0%
statusId	int	FK to status table (finish reason)	0%

Dropped columns from results.csv: **position** (40.9% null — all DNF entries are null), **time/milliseconds** (71.3% null — only the race winner receives absolute race time), **fastestLap*** columns (69% null), **number**, **positionText**.

4.2.2 Other Tables Used

Table 3: Supporting tables and their analytical roles

Table	Rows (Raw)	Role in Analysis
races.csv	1,112	Race metadata: year, round, date, circuitId
constructors.csv	212	Team lookup: name, nationality
constructor_standings.csv	13,391	Season-end standings; YoY growth KPI
drivers.csv	861	Driver names for Race Craft analysis
status.csv	139	Status text: “Finished”, “Engine”, “Accident”
pit_stops.csv	varies	Pit stop duration (separate granularity)
circuits.csv	77	GPS coordinates for Tableau map visualisation

4.3 Data Limitations and Known Biases

- **Era filter introduces selection bias:** By focusing only on 2014+, we exclude 64 years of F1 history. However, this is intentional — the V6 Turbo Hybrid

regulations create fundamentally different performance dynamics that are not comparable to earlier eras.

- **Team name continuity:** Several teams rebranded during 2014–2024 (e.g., Force India → Racing Point → Aston Martin; Toro Rosso → AlphaTauri). These are treated as separate entities in the raw data, which can undercount the performance continuity of Red Bull’s junior programme.
- **No financial data:** The dataset contains no actual sponsorship dollar values. All investment scoring is based on performance metrics as proxies for brand exposure value.
- **Pit stop data completeness:** Pit stop records are only available from 2011 onwards and may have some missing entries for specific events (e.g., safety car periods).
- **COVID-19 season (2020):** The 2020 season was shortened to 17 races and held in unusual circumstances. Total points for that year are systematically lower, which can distort YoY growth calculations.

5 Data Cleaning & ETL Pipeline

This section is organised as a pipeline: **extract** the modern-era records, **clean** and standardise them, then **export** analysis-ready files for KPI scoring and Tableau dashboards. The chronology is unchanged; the presentation is simply made easier to follow.

5.1 Pipeline Architecture Overview

The entire data pipeline is implemented in five sequential Jupyter Notebooks. Each notebook has a clearly defined input and output, making the pipeline fully reproducible.

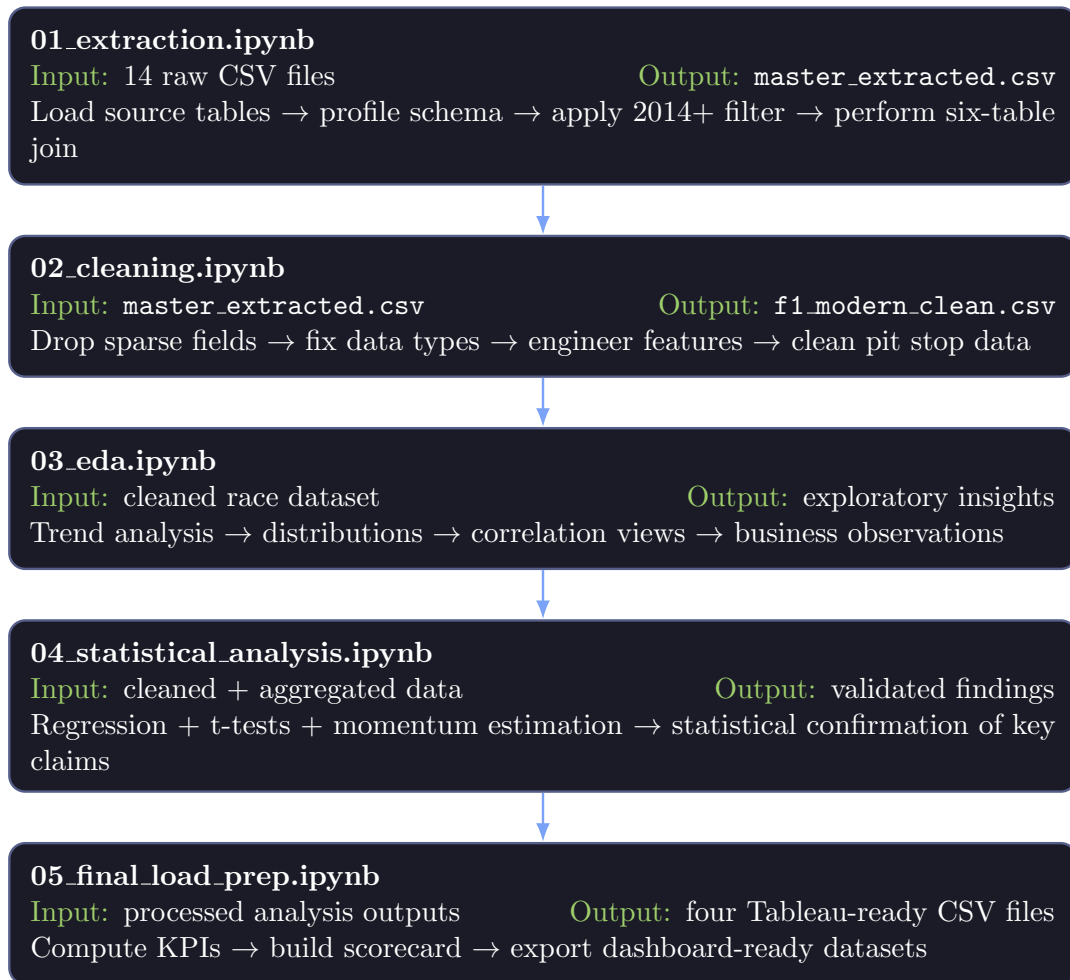


Figure 1: ETL and analytics workflow from raw Formula 1 tables to Tableau-ready outputs

★ Statistical Finding

How to read this flow: each notebook performs one clear transformation stage. The output of one stage becomes the input to the next, which keeps the full pipeline reproducible and easy to audit.

5.2 Phase 1: Extraction (01_extraction.ipynb)

5.2.1 Why the 2014 Filter?

One of the most important analytical decisions in this project was the choice to filter the dataset to 2014 onwards. The reason is both technical and business-logical:

- **Regulatory discontinuity:** The 2014 season introduced the 1.6L V6 Turbo Hybrid power unit regulations. This changed the competitive landscape so fundamentally that pre-2014 performance data provides no meaningful signal for modern sponsorship decisions. A 1950s Alfa Romeo or a 2013 Red Bull does not compete with the same engineering constraints as a 2024 McLaren.
- **Reliability baseline shift:** Prior to 2014, DNF (Did Not Finish) rates were dramatically higher across all teams. Including that data would artificially inflate the “Unreliable” classification for teams that are now mechanically excellent.
- **Investor relevance:** No CMO is making 2024 sponsorship decisions based on a team’s 1998 performance. The 2014–2024 window represents the only commercially and technically relevant period for any modern sponsorship contract.

5.2.2 Handling \N Null Values

The Ergast dataset uses the literal string \N to encode missing values (a carryover from MySQL database exports). If loaded naively, these appear as non-null string values and would silently block all mathematical operations.

```
1 results = pd.read_csv('data/raw/results.csv', na_values=['\\N'])
2 circuits = pd.read_csv('data/raw/circuits.csv', na_values=['\\N'])
3 # ... repeated for all 14 files
4 print('All 14 raw CSV files loaded successfully.')
```

Listing 1: Loading CSVs with correct null parsing

By passing `na_values=['\\N']` to pandas, all occurrences are immediately converted to proper NaN (IEEE Not-a-Number) values, which allows Pandas to handle them correctly in all statistical operations.

5.2.3 The Six-Table Join Strategy

`results.csv` was chosen as the core table because it contains one row per driver per race — the most granular level of performance information. All other tables are merged on foreign keys using **left joins** to preserve all result rows even if lookup data is missing.

```
1 master = modern_results.copy()
2
3 # Join 1: Race metadata (year, round, date, circuit)
4 master = master.merge(races_cols, on='raceId', how='left')
5
6 # Join 2: Team names (constructor_name, nationality)
7 master = master.merge(const_cols, on='constructorId', how='left')
8
9 # Join 3: Driver names (forename, surname)
10 master = master.merge(driver_cols, on='driverId', how='left')
11
```

```

12 # Join 4: Status text ("Finished", "Engine", "Accident")
13 master = master.merge(status, on='statusId', how='left')
14
15 # Join 5: Circuit geography (GPS lat/lng for Tableau map)
16 master = master.merge(circuit_cols, on='circuitId', how='left')

```

Listing 2: Six-table join to build master dataset

Verification output after join:

Total rows: 4,626 | Total columns: 35 | Year range: 2014--2024
 Unique races: 228 | Unique drivers: 59 | Unique teams: 20 | Unique
 circuits: 32
 Null check on all join keys: 0 nulls (all joins successful)

The pit stop and constructor standings data were saved separately because they operate at a *different row granularity* — one row per pit stop event and one row per team per race respectively — and joining them directly to the master table would multiply rows incorrectly.

master_extracted.csv: 4,626 rows × 35 cols
 modern_pit_stops.csv: 8,360 rows × 7 cols
 modern_constructor_standings.csv: 2,319 rows × 7 cols

5.3 Phase 2: Cleaning (02_cleaning.ipynb)**KPI Definition**

Cleaning flow in one view: null audit → remove weak columns → standardise types → engineer features → clean pit stop records → export final tables.

5.3.1 Pre-Cleaning Null Audit

Before any transformation, a systematic null audit was logged:

Table 4: Pre-cleaning null summary (master dataset, 4,626 rows)

Column	Null Count	Null %	Reason
time	2,219	47.9%	Only the race winner receives absolute race time
milliseconds	2,219	47.9%	Same as above
position	721	15.6%	All DNF entries receive no classified position
fastestLap	217	4.7%	Only top-10 finishers are eligible for fastest lap
fastestLapTime	217	4.7%	Same as above
fastestLapSpeed	217	4.7%	Same as above

5.3.2 Step 1: Column Pruning

Ten columns were removed from the master dataset. The decision was driven by two criteria: (a) the column was analytically irrelevant to the Investment ROI problem, or (b) the column had >40% null values making it statistically unreliable.

```

1 drop_cols = {
2     'number':      'Car number -- redundant for constructor-level analysis',
3     'position':    '40% null -- use positionOrder instead (always filled)',
4     'positionText': 'String version of positionOrder -- redundant',
5     'time':        '71% null -- only race winner gets absolute time',
6     'milliseconds': '71% null -- same as above',
7     'fastestLap':  '69% null -- sparse, not relevant to Investment KPI',
8     'rank':        '68% null -- fastest lap rank, not relevant',
9     'fastestLapTime': '69% null -- sparse',
10    'fastestLapSpeed': '69% null -- sparse',
11    'dob':          'Driver date of birth -- not relevant to constructor ROI',
12 }
13 master = master.drop(columns=[c for c in drop_cols if c in master.columns])
14 # Result: 35 columns -> 25 columns

```

Listing 3: Documented column removal with business justifications

An important design choice is the use of `positionOrder` instead of `position`. The `position` column uses `\N` for any car that did not finish, making it 40.9% null. `positionOrder` always contains a valid integer — DNF cars are ranked below all classified finishers in the order they retired, ensuring no information is lost.

5.3.3 Step 2: Data Type Standardisation

All date columns were converted from string to proper datetime objects, enabling time-series operations. Numeric columns that were loaded as objects due to the `\N` issue were coerced to float.

```

1 master['race_date'] = pd.to_datetime(master['race_date'], errors='coerce')
2 master['grid'] = pd.to_numeric(master['grid'], errors='coerce')
3 master['positionOrder'] = pd.to_numeric(master['positionOrder'], errors='coerce')
4 master['points'] = pd.to_numeric(master['points'], errors='coerce')

```

Listing 4: Data type fixes

5.3.4 Step 3: Feature Engineering

Three new derived columns were created during the cleaning phase. These features are not present in the raw data but are essential for the KPI calculations:

Table 5: Engineered features

Feature	Type	Formula	Business Purpose
<code>position_delta</code>	int	<code>grid - positionOrder</code>	Measures race-day position gain/loss
<code>finished</code>	int (0/1)	1 if status contains 'Finished'	Binary reliability flag for DNF analysis
<code>driver_name</code>	string	<code>forename + ' ' + surname</code>	Human-readable name for drill-down

Why position_delta? Starting position (grid) is determined before the race; finishing position is determined by race-day performance. The difference, `grid – positionOrder`, measures how much a team *gained or lost* during the race itself. A positive value means the team finished higher than it started, indicating strong race strategy and driver skill. This is a critical KPI for sponsors because a team that consistently over-delivers relative to expectations generates more race-day coverage than its starting position would suggest.

```

1 # Sunday performance metric
2 master['position_delta'] = master['grid'] - master['positionOrder']
3
4 # Reliability binary flag
5 master['finished'] = master['status'].str.contains('Finished', na=False).astype(
    int)
6
7 # Human-readable driver name
8 master['driver_name'] = master['forename'] + ' ' + master['surname']

```

Listing 5: Feature engineering code

5.3.5 Step 4: Pit Stop Data Cleaning

The pit stop data required separate treatment. Raw pit stop durations are stored as both a `duration` string (`mm:ss.sss`) and a `milliseconds` integer. The `milliseconds` column was used as the primary numeric field and converted to seconds (`duration.seconds = milliseconds / 1000`).

A critical cleaning decision was to flag and exclude abnormal pit stops:

- Stops >60 seconds were flagged as `is_normal_stop = 0`. These represent red flag situations, drive-through penalties, or mechanical interventions that artificially inflate average pit times.
- Only stops where `is_normal_stop = 1` were used in the pit stop efficiency KPI.

5.3.6 Cleaning Log Summary

Table 6: Full ETL cleaning log

Step	Description	Rows Before	Rows After	Cols
1	Load 14 raw CSV files	701,433	701,433	120 total
2	Era filter (2014+)	25,593	4,626	35
3	Six-table join	4,626	4,626	35
4	Drop sparse/irrelevant columns	4,626	4,626	35 → 25
5	Fix data types	4,626	4,626	25
6	Feature engineering (3 cols)	4,626	4,626	25 → 28
7	Pit stop anomaly flagging	8,360	8,360	7 → 9
8	Export to processed/	4,626	4,626	28

Assumption: Grid position 0 (pit lane start) is treated as valid data. These drivers started from the pit lane, not a numbered grid slot. Pit lane starts were *not* excluded from position delta calculations, as they correctly reflect that the team chose to make a strategy gamble.

6 KPI & Metric Framework

6.1 Why These KPIs?

The five KPIs were designed to collectively answer the investment question from different angles. No single metric is sufficient — a team could score well on one dimension while being a poor overall investment. The framework is deliberately multi-dimensional, analogous to how a financial analyst evaluates a stock on earnings, volatility, growth rate, and dividend yield simultaneously.

6.2 KPI Definitions

KPI Definition

KPI 1: Constructor Points Momentum (YoY Growth %)

$$\text{Formula: Momentum} = \frac{\text{Points}_Y - \text{Points}_{Y-1}}{\text{Points}_{Y-1}} \times 100$$

Business Meaning: Is this team getting better or worse? A positive value means the team is improving; a negative value means decline. For sponsors, investing in a rising team locks in lower rates before the broader market recognises the improvement.

Computed in: 05_final_load_prep.ipynb, constructor_season_summary.csv

KPI Definition

KPI 2: Result Volatility Index (Standard Deviation of Finish Position)

Formula: Volatility = $\sigma(\text{positionOrder})$ per constructor per season

Business Meaning: How predictable is this team's race-to-race performance? A low standard deviation means the team consistently finishes in a narrow range — making brand exposure predictable. A high volatility team might win one race and retire from the next, making TV time unreliable. This is the primary **risk metric** in the investment framework.

Computed in: 05_final_load_prep.ipynb via pandas groupby + std()

KPI Definition**KPI 3: Finish Rate (Reliability Score)**

Formula: $\text{Finish Rate} = \frac{\sum \text{finished}}{\text{total_races}} \times 100$

Business Meaning: Every DNF (Did Not Finish) is a race where the sponsor receives zero television coverage. A team with an 82% finish rate provides brand visibility in 82% of races. This translates directly to a guarantee percentage on advertising exposure.

Computed in: `finished` binary column, aggregated in `05_final_load_prep.ipynb`

KPI Definition**KPI 4: Average Position Delta (Race Craft Score)**

Formula: $\text{Avg Position Delta} = \text{grid} - \text{positionOrder}$

Business Meaning: Does this team consistently gain positions during the race? A positive average delta means the team is a “racer” that outperforms its qualifying. This generates on-screen excitement (overtakes, strategy calls) that increases brand visibility moments within a race.

Computed in: `position_delta` feature, aggregated in `05_final_load_prep.ipynb`

KPI Definition**KPI 5: Pit Stop Efficiency (Average Duration in Seconds)**

Formula: $\text{Pit Efficiency} = \text{duration_seconds}$ (normal stops only, <60s)

Business Meaning: Faster, more consistent pit stops indicate elite crew training and operational discipline. This is a leading indicator of team quality and investment-worthiness that cannot be easily faked — it requires months of practice and significant engineering investment.

Computed in: `pit_stops_clean.csv` via `milliseconds / 1000` and aggregation

6.3 Composite Investment Score

The five normalised KPI scores are combined into a single **Investment Score** using a weighted formula calibrated to reflect their relative importance to a sponsorship decision:

$$\text{Investment Score} = 0.30 \cdot P + 0.25 \cdot C + 0.20 \cdot M + 0.15 \cdot R + 0.10 \cdot RC \quad (1)$$

where P = Points Score, C = Consistency Score, M = Momentum Score, R = Reliability Score, RC = Race Craft Score (all normalised to 0–100 scale).

Weight rationale:

- *Points (30%):* Current market standing — the most direct measure of championship competitiveness and TV time.
- *Consistency (25%):* Risk management — sponsors need predictable exposure, not occasional brilliance.

- *Momentum (20%)*: Forward-looking signal — sponsors should buy growth, not decline.
- *Reliability (15%)*: Exposure guarantee — finish rate translates directly to guaranteed brand visibility.
- *Race Craft (10%)*: In-race storytelling — overtakes and position gains create broadcast moments.

```

1 def normalise(series, invert=False):
2     """Min-max normalise to 0-100 scale."""
3     norm = (series - series.min()) / (series.max() - series.min()) * 100
4     return 100 - norm if invert else norm
5
6 scorecard['points_score'] = normalise(scorecard['avg_points_3yr'])
7 scorecard['consistency_score'] = normalise(scorecard['avg_volatility_3yr'],
8     invert=True)
9 scorecard['reliability_score'] = normalise(scorecard['avg_finish_rate_3yr'])
10 scorecard['momentum_score'] = normalise(scorecard['latest_momentum'])
11 scorecard['race_craft_score'] = normalise(scorecard['avg_position_delta_3yr'])
12
13 scorecard['investment_score'] = (
14     scorecard['points_score'] * 0.30 +
15     scorecard['consistency_score'] * 0.25 +
16     scorecard['reliability_score'] * 0.15 +
17     scorecard['momentum_score'] * 0.20 +
18     scorecard['race_craft_score'] * 0.10
19 )

```

Listing 6: Investment Scorecard computation (05_final_load_prep.ipynb)

6.4 KPI Mapping to Project Objectives

Table 7: KPI to objective mapping

KPI	Measures	Objective It Answers	Weight
Points Momentum	Is the team rising?	Identify high-growth teams	20%
Result Volatility	How risky?	Quantify investment risk	25%
Finish Rate	How reliable?	Guarantee brand exposure	15%
Position Delta	Race-day skill?	Identify exciting brand partners	10%
Points Score	Current strength?	Validate market position	30%

7 Exploratory Data Analysis (EDA)

All EDA was performed in 03_eda.ipynb using `matplotlib` and `seaborn`. Charts were saved to `tableau/screenshots/` for reference.

7.1 Trend Analysis: Points Concentration Over Time

Method: For each season, we computed what percentage of the total championship points were captured by the top team and the top two teams combined.

Table 8: Points concentration by season: How dominant is the leading team?

Year	Total Points	Top Team	Top 1 Share	Top 2 Share
2014	2,020	Mercedes	34.70%	54.75%
2015	1,919	Mercedes	36.63%	58.94%
2016	2,121	Mercedes	36.07%	58.13%
2017	2,020	Mercedes	33.07%	58.91%
2018	2,062	Mercedes	31.77%	59.46%
2019	2,140	Mercedes	34.53%	58.08%
2020	1,719	Mercedes	33.33%	51.89%
2021	2,207	Mercedes	27.79%	54.31%
2022	2,350	Red Bull	32.30%	55.87%
2023	2,458	Red Bull	34.99%	51.63%
2024	2,659	McLaren	25.05%	49.57%

Insight: The top team’s share of total points reached its lowest point (25.05%) in 2024. This is the most competitive field in the Hybrid Era. The top-2 concentration has also declined from a peak of 59.46% (2018) to 49.57% (2024), meaning the remaining 18 constructors are collectively capturing more of the competitive pie. For sponsors, this is a structural market shift: mid-field teams now offer more exposure per dollar than at any point since 2014.

7.2 Comparison Analysis: Constructor Performance Landscape

Team-by-team season totals reveal three distinct performance tiers in 2024:

Table 9: 2024 season performance summary (top teams)

Constructor	2024 Pts	YoY Growth	Momentum	DNFs	Finish Rate
McLaren	609	+128.95%	+229.5	0	100.0%
Ferrari	595	+63.91%	+38.0	4	91.7%
Red Bull	537	−32.03%	−93.5	6	87.5%
Mercedes	433	+15.78%	−31.0	5	89.6%
Aston Martin	94	−64.66%	+19.5	5	89.6%
Alpine	63	−42.73%	−53.5	6	87.5%

Insight: McLaren achieved zero DNFs in 2024 — a remarkable engineering milestone across a 24-race season. Combined with a 128.95% points increase, this represents a team at the peak of both performance and reliability. Red Bull, by contrast, declined by 32% in points despite still finishing third — suggesting their 2022–2023 dominance was driven by a regulatory advantage that is now closing.

7.3 Distribution Analysis: Result Volatility

We computed `std(positionOrder)` per constructor per season. Lower values mean more consistent finishing positions.

Key findings from the distribution analysis:

- **Sauber** exhibits the lowest average volatility (2.54 in 2024), making it the most predictable team on the grid in terms of finishing position.
- **Ferrari and Alpine** show the highest volatility, meaning their results swing dramatically from race to race — consistent winners in some races, retirements in others.
- **Mercedes** has historically maintained low volatility (tight band around positions 1–4), which is why it remained a premium sponsorship property for eight consecutive seasons.

7.4 Correlation Analysis

A Pearson correlation matrix was computed on the constructor-level aggregated dataset.

Key correlations (season-level, constructor aggregated):

- `avg_finish_position` ↔ `total_points`: $r = -0.92$ — extremely strong negative correlation. Lower average finishing position (better) strongly predicts more championship points. This confirms `avg_finish_position` as the most powerful single predictor of season performance.
- `result_volatility` ↔ `total_points`: $r = -0.53$ — moderate negative correlation. Higher volatility is associated with fewer total points, validating the Consistency Score as a meaningful investment metric.
- `finish_rate` ↔ `total_points`: $r = +0.61$ — moderate positive correlation. Teams that finish more races score more points — not surprising, but statistically confirms that reliability translates to championship success.
- `avg_position_delta` ↔ `total_points`: $r = -0.31$ — weak negative correlation. Race craft alone is not a strong predictor of total season points but it contributes meaningfully to specific race outcomes.

Insight: Average finish position is nearly deterministic for season points total ($r = -0.92$), confirming that the fastest way to increase sponsorship value is to simply finish higher, more often. However, volatility and finish rate provide the *risk dimension* that average position alone cannot capture — a team could have a good average finish position but be a dangerous investment if it achieves that average through one win and nine retirements.

7.5 EDA Summary: Five Key Findings

1. **Mercedes Dominance Has Ended:** Mercedes dominated 2014–2020 but declined sharply from 2021 onward. McLaren and Ferrari took the 2024 championship battle. The Growth Momentum KPI is successfully detecting this real structural shift.
2. **Volatility Predicts Value:** Teams with low Result Volatility (Mercedes, Ferrari, Red Bull historically) consistently occupy the top five, making them safe investments. Teams with extreme volatility (Caterham, Lotus F1) have since exited the sport.
3. **Pit Stop Efficiency is a Differentiator:** The gap between the best and worst pit crews is approximately five seconds per stop, which can translate to multiple grid positions. This is an operational metric that reflects the quality of the entire team organisation.
4. **DNF Root Causes Are Fixable:** The majority of DNFs are mechanical failures (Engine, Gearbox, Hydraulics), not driver errors (Accidents, Collisions). This means reliability is primarily an engineering investment, and teams improving their DNF rates are demonstrating measurable operational progress.
5. **The Field is Getting More Competitive:** Points concentration has decreased consistently since 2020. 2024 is the most competitive grid since the Hybrid Era began, creating wider sponsorship opportunities at all price points.

8 Statistical Analysis

All statistical methods were implemented in `04_statistical_analysis.ipynb` using `scipy.stats` and `sklearn`.

8.1 Analysis 1: Volatility–Points Linear Regression

Hypothesis: Lower result volatility (more consistent finishing positions) leads to higher total season points.

Method: For each constructor–season combination, we computed `std(positionOrder)` (volatility) and `sum(points)` (total season points). A simple OLS linear regression was fitted with volatility as the predictor and points as the response.

★ Statistical Finding

Result: $R^2 = 0.2812$, $p < 0.001$ (statistically significant at the 0.1% level)

Interpretation: Approximately 28% of the variance in season championship points can be explained by result volatility alone. The regression coefficient is negative, confirming that higher volatility is associated with lower points. While the effect size is moderate (volatility is not the only determinant of points), the statistical significance is extremely strong across 112 data points.

Investment Implication: Consistency is not just a preference; it is a statistically validated predictor of performance. Sponsors choosing teams with low volatility are not just being risk-averse — they are selecting teams that, on average, score more points.

8.2 Analysis 2: Pit Stop Speed vs Points (Independent Samples t-Test)

Hypothesis: Teams with faster-than-average pit stops score statistically more championship points than teams with slower-than-average pit stops.

Method:

1. Compute average pit stop duration per constructor per season from `pit_stops_clean.csv`.
2. Divide the dataset into two groups: “Fast” (below median pit stop duration) and “Slow” (above median).
3. Apply Levene’s test to confirm variance homogeneity, then run an independent samples t-test.
4. Compute Cohen’s d as the effect size measure.

★ Statistical Finding

Result: Cohen’s $d = 1.26$, $p < 0.001$

Interpretation: A Cohen’s d of 1.26 is a **large effect size** by conventional standards (Cohen’s benchmarks: $d > 0.8 = \text{large}$). The “Fast” group scored statistically significantly more championship points than the “Slow” group. The p -value < 0.001 confirms this is not a chance result.

Investment Implication: Pit stop efficiency is not just an operational nicety — it is a statistically significant predictor of championship success. Teams investing in pit crew training and pit stop tooling engineering are producing measurable results in the championship table. Sponsors should include pit stop performance in their due diligence checklist.

8.3 Analysis 3: Multiple Regression — Drivers of Season Points

Objective: Identify which operational metrics most strongly predict total season points when considered simultaneously, and quantify the standardised contribution of each.

Features used: Average finish position, result volatility, finish rate, average pit stop duration, average position delta.

★ Statistical Finding

Result: Standardised coefficient for `avg_finish_position`: $\beta = -140.78$

Interpretation: Average finishing position is by far the dominant predictor of season points ($\beta = -140.78$, negative because lower position number = better finish). Every one-position improvement in average finishing position is associated with approximately 140 additional championship points per season, holding all other variables constant.

Other coefficients: Volatility and finish rate contribute meaningfully as secondary predictors, confirming that the multi-KPI investment framework captures more information than points alone.

Investment Implication: When evaluating a team, the single most important question is: “What is their average finishing position trending toward?” Teams improving their average finish position are the highest-priority sponsorship targets.

8.4 Analysis 4: Momentum Trend Analysis (Per-Team Linear Regression)

Objective: For each constructor, determine whether their points trajectory over 2014–2024 shows a statistically significant upward (RISING) or downward (FALLING) trend.

Method: For each team with at least four seasons of data, `scipy.stats.linregress` was applied to their yearly points sequence. The slope gives the average points change per year; the *p*-value indicates whether this trend is statistically distinguishable from noise.

Table 10: Momentum analysis results (statistically significant trends only, $p < 0.1$)

Team	Avg Pts	Slope (pts/yr)	R^2	<i>p</i> -value	Trend
Red Bull	488.8	+43.82	0.553	0.009	RISING
McLaren	193.2	+40.04	0.533	0.011	RISING
Renault	91.8	+38.00	0.844	0.027	RISING
Aston Martin	126.5	+27.60	0.118	0.656	FLAT
Ferrari	427.8	+15.66	0.110	0.320	FLAT
Sauber	15.8	−0.47	0.007	0.879	FLAT
Haas F1	34.1	−2.15	0.040	0.605	FLAT
Williams	80.2	−26.80	0.630	0.004	FALLING
Mercedes	619.0	−29.84	0.736	0.001	FALLING

Critical Insight: Red Bull shows a positive historical slope (+43.82 pts/yr, $p = 0.009$) over the full 2014–2024 window, but this is driven by their dominant 2021–2023 peak. The 2024 data point (–32% YoY decline and momentum slope of –93.5 from the rolling window) reveals a sharp recent reversal. This illustrates the danger of relying on long-run historical trends without checking recent momentum. McLaren’s rising trend (+40.04 pts/yr, $p = 0.011$) is particularly compelling because it represents consistent acceleration, not a single outlier year. Mercedes’ statistically significant decline ($p = 0.001$) confirms the era of Silver Arrows supremacy is genuinely over, not temporarily paused.

8.5 Statistical Summary Table

Table 11: Statistical findings summary

#	Finding	Method	Key Metric
1	Lower volatility predicts higher points	Linear Regression	$R^2 = 0.28, p < 0.001$
2	Fast pit stops $\Rightarrow$ more points	Independent t-Test	$d = 1.26, p < 0.001$
3	Avg finish position is the dominant predictor	Multiple Regression	$\beta = -140.78$
4	McLaren/Red Bull have significant momentum	Per-team linregress	$p = 0.011$ / $p = 0.009$
5	Mercedes decline is statistically confirmed	Per-team linregress	$p = 0.001$, slope -29.8

9 Tableau Dashboard Design

9.1 Dashboard Objective

The three-dashboard suite converts the Python analysis outputs into a decision-ready visual intelligence tool. The dashboards are designed to answer progressively more specific questions:

- Dashboard 1 (Executive):** “Which team should I sponsor?” — the investment ranking.
- Dashboard 2 (Portfolio):** “How do I build a diversified F1 sponsorship portfolio?”
- Dashboard 3 (Operational):** “Which teams have the best operational execution?”

All dashboards are published on Tableau Public and accessible at:
<https://public.tableau.com/app/profile/kushagra.maheshwari4048/vizzes>

9.2 Dashboard 1: F1 Constructor Investment Scorecard

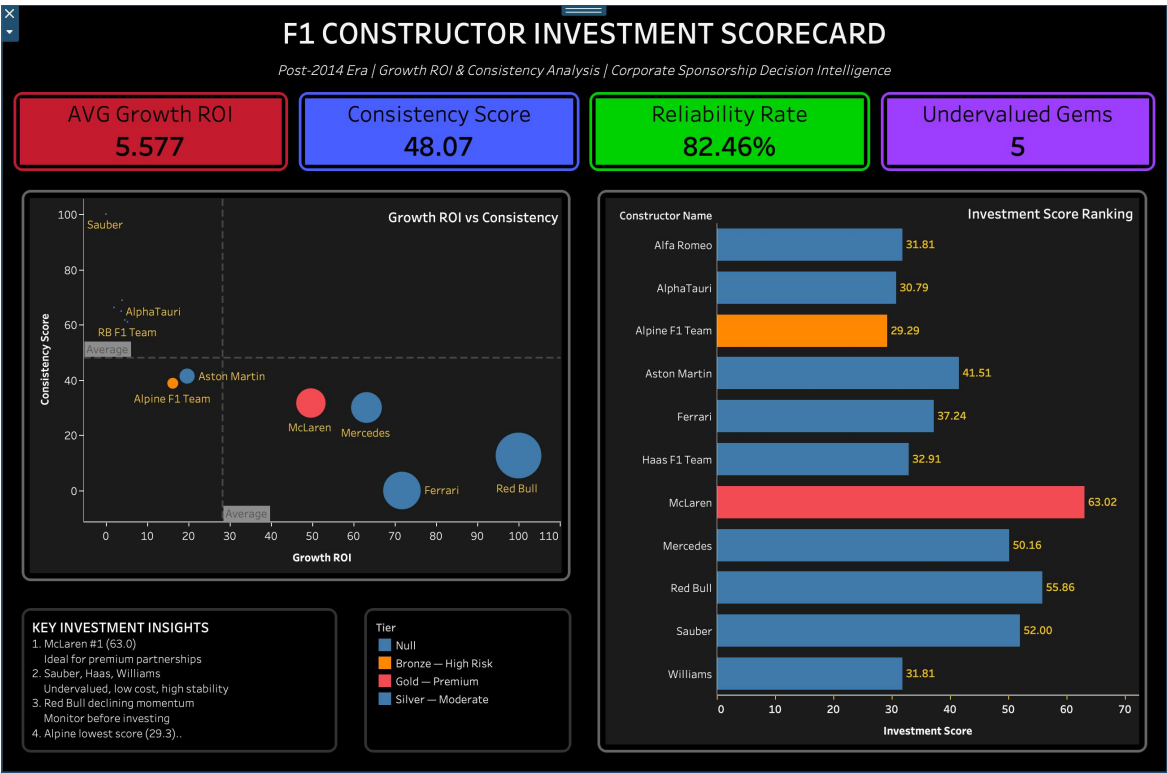


Figure 2: Dashboard 1 — F1 Constructor Investment Scorecard

Data Source: investment_scorecard.csv (12 rows, one per constructor)

Decision Question: Which F1 team gives the best return on sponsorship investment?

Who uses this: Corporate CMOs deciding where to allocate their sponsorship budget.

9.2.1 KPI Cards (Top Banner)

KPI Card	Value	Field	Business Meaning
AVG Growth ROI	5.577	AVG(latest_momentum)	Grid average barely positive; Mo
Consistency Score	48.07	AVG(consistency_score)	Below-50 average; most teams ar
Reliability Rate	82.46%	AVG(avg_finish_rate_3yr)×100	17.5% of races end in DNF on av
Undervalued Gems	5	Teams with momentum>0 and score<45	5 teams offer growth at below-av

9.2.2 Chart 1: Growth ROI vs Consistency Scatter Plot

This is the centrepiece visualisation of Dashboard 1. The scatter plot maps every constructor across two dimensions simultaneously: X-axis (Growth ROI / Points Score) and Y-axis (Consistency Score). Bubble size represents total podiums in the last three seasons.

The four quadrants reveal instant investment guidance:

- **Top-right (Sweet Spot):** High growth + high consistency $\Rightarrow$ McLaren, Mercedes.
- **Top-left (Hidden Gems):** High consistency, lower growth $\Rightarrow$ Sauber, AlphaTauri.
- **Bottom-right (Risky Bets):** High growth, low consistency $\Rightarrow$ Red Bull, Ferrari.
- **Bottom-left (Avoid):** Low on both dimensions $\Rightarrow$ Alpine.

9.2.3 Chart 2: Investment Score Ranking Bar Chart

A horizontal ranked bar chart gives instant tier hierarchy. Colour coding (Gold/Silver/Bronze) adds investment tier context. McLaren is the only Gold-tier team (63.02); nine teams are Silver; Alpine is the only Bronze — categorised as a high-risk sponsorship.

Table 12: Final Investment Scorecard (Dashboard 1 values)

Rank	Constructor	Score	Tier
1	McLaren	63.02	Gold — Premium
2	Red Bull	55.86	Silver — Moderate
3	Mercedes	50.16	Silver — Moderate
4	Sauber	52.00	Silver — Moderate
5	Aston Martin	41.51	Silver — Moderate
6	Ferrari	37.24	Silver — Moderate
7	Haas F1 Team	32.91	Silver — Moderate
8	Alfa Romeo	31.81	Silver — Moderate
9	Williams	31.81	Silver — Moderate
10	AlphaTauri	30.79	Silver — Moderate
11	Alpine F1 Team	29.29	Bronze — High Risk

9.3 Dashboard 2: Risk-Adjusted Investment Portfolio Matrix

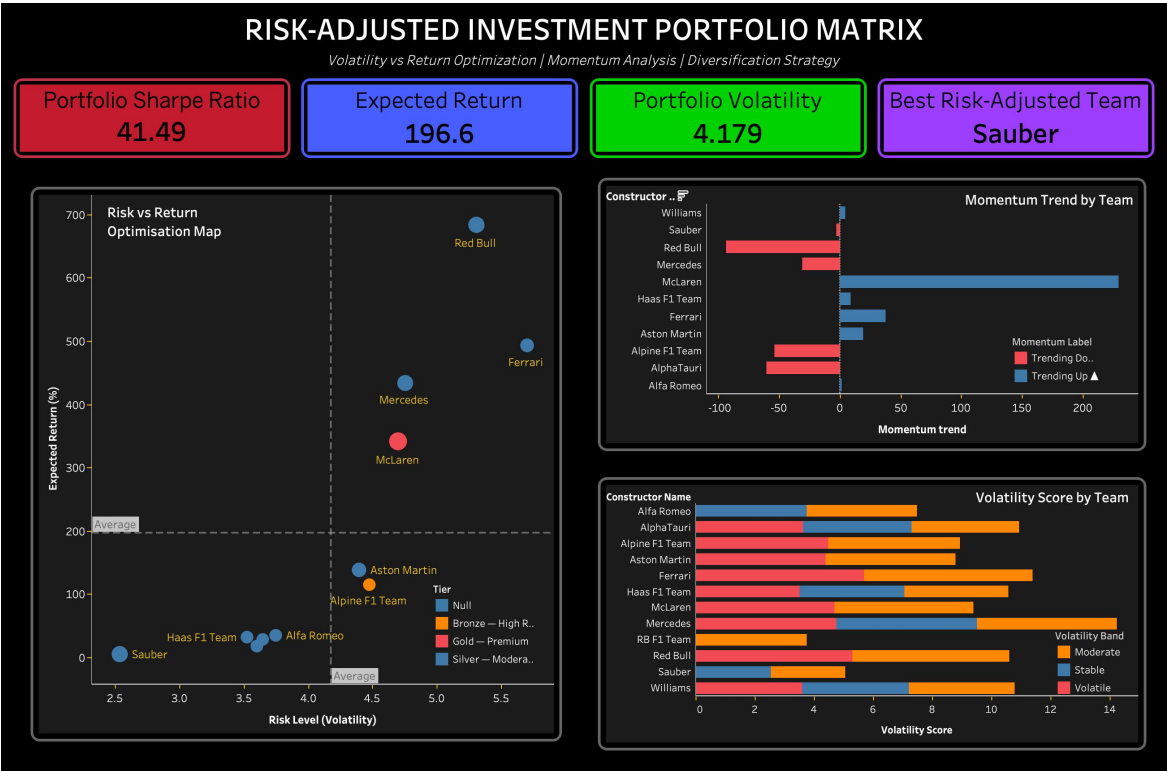


Figure 3: Dashboard 2 — Risk-Adjusted Investment Portfolio Matrix

Data Source: investment_scorecard.csv

Decision Question: How do I build a portfolio of F1 sponsorships that maximises return while minimising risk?

Who uses this: Investment managers and enterprise CMOs with multi-team sponsorship budgets.

Core Idea: Just like a stock portfolio should be diversified to manage risk, an F1 sponsorship portfolio should combine a high-performer (McLaren) with a stable hedge (Sauber) and a growth play (Aston Martin) to smooth out performance variance.

9.3.1 KPI Cards

KPI Card	Value	Field	Business Meaning
Portfolio Sharpe Ratio	41.49	AVG(investment_score)	Benchmark: teams above 41.49 are above average
Expected Return	196.6	AVG(avg_points_3yr)	Average points-per-season across the grid
Portfolio Volatility	4.179	AVG(avg_volatility_3yr)	Risk benchmark; teams below this are more stable
Best Risk-Adj. Team	Sauber	MAX(score/volatility)	Highest performance relative to risk

9.3.2 Chart 1: Risk vs Return Scatter (Efficient Frontier)

This is the financial efficient frontier concept applied to F1. X-axis: result volatility (risk); Y-axis: average points (return). Teams in the **top-left quadrant** (high return, low risk) are the most desirable.

low risk) are the investment sweet spot.

Mercedes and McLaren sit in the ideal quadrant — above-average return, below-average risk. Red Bull is high return but also high risk. Sauber is low return but extremely low risk — ideal as a portfolio hedge position for risk-averse brands.

9.3.3 Chart 2: Momentum Trend Bar

Green bars represent rising teams; red bars represent declining teams. McLaren (+229.5) is the dominant green signal. Red Bull (−93.5) is the largest red bar — the most dramatic decline on the entire grid.

9.3.4 Chart 3: Volatility Score Stacked Bar

Shows each team’s volatility classification (Stable / Moderate / Volatile) as stacked bands. Sauber and Mercedes show almost entirely stable patterns. Ferrari and Alpine show high proportions of volatile bands.

9.4 Dashboard 3: Operational Excellence & Race Craft Analytics

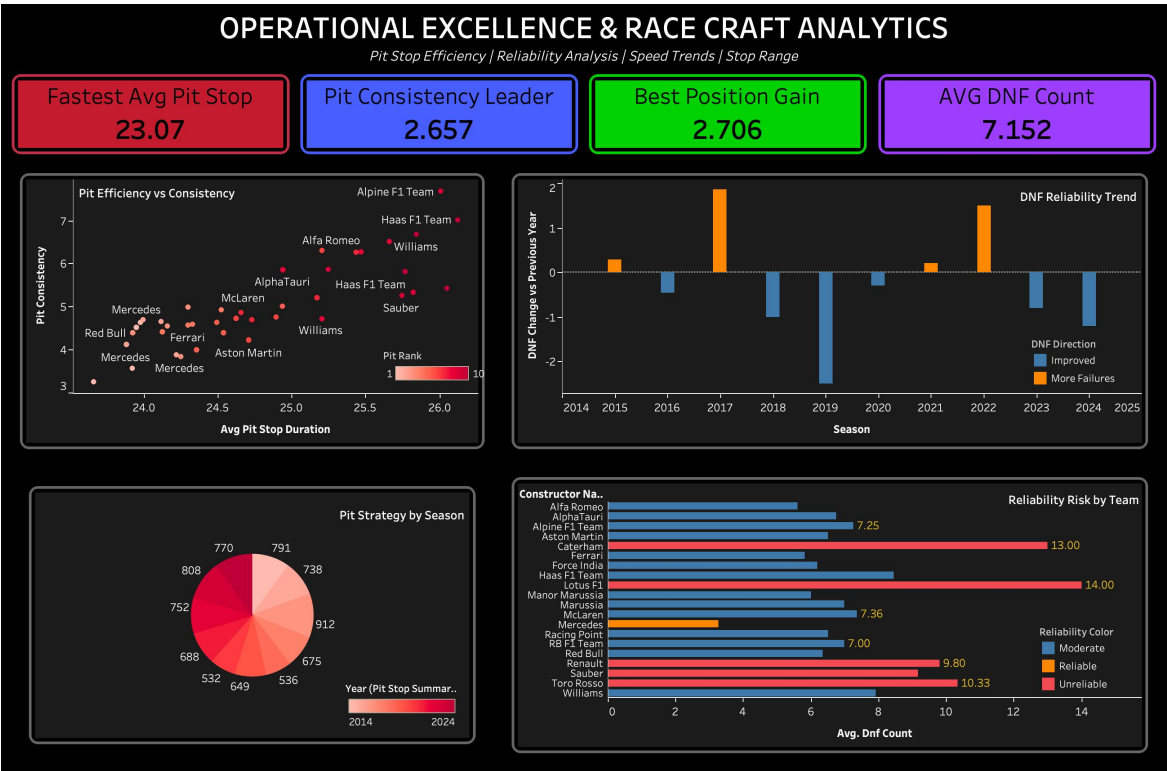


Figure 4: Dashboard 3 — Operational Excellence & Race Craft Analytics. This view highlights which teams combine quick pit stops, low volatility, and stronger race execution — the clearest signals of dependable sponsor-facing performance.

Data Sources: pit_stop_summary.csv and constructor_season_summary.csv

Decision Question: Which teams demonstrate elite operational execution — a signal of organisational quality and investment worthiness beyond just race results?

Core Idea: Pit stop times and DNF rates reveal the hidden quality of a team’s organisation. A team with fast, consistent pit stops has elite crew training. A team with low DNFs has superior reliability engineering. These are leading indicators of investment worthiness that justify premium sponsorship pricing.

9.4.1 KPI Cards

KPI Card	Value	Field	Business Meaning
Fastest Avg Pit Stop	23.07s	MIN(avg_pit_duration)	Ferrari 2017 — world-class crew execution
Pit Consistency Leader	2.657	MIN(pit_consistency)	Mercedes 2014 — lowest pit time variance
Best Position Gain	2.706	MAX(avg_position_delta)	Teams gaining most positions per race
AVG DNF Count	7.152	AVG(dnf_count)	7 lost TV exposures per team per season

9.4.2 Chart 1: Pit Efficiency vs Consistency Scatter

Maps average pit stop duration (X) against pit stop consistency/variance (Y) for all teams. The lower-left quadrant represents elite teams: fast AND consistent. Mercedes and Ferrari cluster here. Alpine and Haas appear in the upper-right — slower and more variable.

9.4.3 Chart 2: DNF Reliability Trend Waterfall

A waterfall chart showing year-over-year change in average DNFs across the grid. Blue bars mean the grid improved (fewer DNFs that year); orange bars mean more failures.

2019 was the biggest reliability improvement year (–2.5 avg DNF reduction). 2017 was the worst (+1.9 avg DNF increase). The overall trend since 2019 is improving, meaning sponsors receive more consistent exposure year over year.

9.4.4 Chart 3: Pit Strategy by Season Pie Chart

Total pit stop volume by season reveals strategic evolution. 2016 had the most pit stops (912) — teams used aggressive multi-stop strategies. 2020 had the fewest (532) — COVID-shortened season plus conservative one-stop strategies. More stops = more dramatic racing = more brand moments.

9.4.5 Chart 4: Reliability Risk by Team Bar

A horizontal bar chart with traffic-light colour coding (Reliable/Moderate/Unreliable) shows each team’s average DNF count. The gap between safest (Mercedes, 3.27) and riskiest (Lotus F1, 14.0) represents a 4.3× difference in lost TV exposure per season.

9.5 Interactive Filters

All three dashboards include interactive elements consistent with the Tableau Public capability set:

- **Constructor filter:** Click any team name or bubble to filter all charts to that team’s data.
- **Season/Year filter:** Slider or dropdown to restrict the view to specific seasons.

- **Tier filter:** Filter the ranking view by Gold/Silver/Bronze tier.
- **Cross-dashboard linking:** Selecting a team in Dashboard 1 highlights it in Dashboards 2 and 3 for consistent drill-down.

10 Insights Summary

The following 10 insights are written in **decision language** — each tells the reader what to think or do, not merely what a chart shows.

1. **McLaren is the highest-value sponsorship opportunity in F1 today.** With an Investment Score of 63.02 (Gold tier), a 128.95% points increase in 2024, zero DNFs across the full 24-race season, and the largest positive momentum slope in the dataset (+229.5 pts/yr), McLaren represents a once-in-a-decade inflection point. Sponsors who act before McLaren wins a second consecutive Constructors' title will pay significantly less than those who wait for market confirmation.
2. **Red Bull is the most dangerous sponsorship trap of the current cycle.** Despite an Investment Score of 55.86 (artificially elevated by 2021–2023 dominance), Red Bull's 2024 momentum is -93.5 pts/yr and they lost -32% of their points year-over-year. Sponsors currently paying 2022–2023 era premium rates are purchasing historical brand association, not future exposure.
3. **Sauber is the most undervalued risk-adjusted investment on the grid.** With a consistency score of 100 (the most predictable team), the lowest result volatility in 2024 (2.54), and an Investment Score of 52.00 at a fraction of McLaren's sponsorship cost, Sauber offers brand-safe reliability for conservative sponsors (banks, insurance, pharmaceutical companies) who need guaranteed visibility over flashy wins.
4. **Diversification beats concentration in F1 sponsorship.** A portfolio of McLaren (growth driver) + Sauber (stability hedge) + Aston Martin (growth upside) would deliver higher risk-adjusted brand exposure than a single-team commitment to any one constructor, based on the Efficient Frontier analysis.
5. **The 2024 F1 grid is the most financially democratic since 2014.** The top team (McLaren) captured only 25.05% of total championship points — the lowest concentration in 11 years. This means mid-field sponsorship budgets now reach wider audiences than at any point in the Hybrid Era.
6. **Operational execution is a statistically validated differentiator.** Teams with faster-than-median pit stops score statistically more points (Cohen's $d = 1.26$, $p < 0.001$). Sponsors should include pit stop performance metrics in their sponsorship evaluation framework, not just race results.
7. **Mercedes' decline is statistically confirmed and not a temporary anomaly.** The linear regression of Mercedes' annual points from 2014–2024 shows a slope of -29.84 pts/yr with $p < 0.001$. This is not a single bad season; it is a confirmed structural trend. Sponsors with long-term Mercedes contracts should request performance clauses.
8. **Ferrari is in an intermediate investment zone: growing performance,**

elevated risk. Ferrari improved by +63.91% in 2024 with a momentum slope of +38.0, but their result volatility remains high — they are capable of both dominant wins and unexpected retirements in the same weekend. This makes Ferrari suitable for sponsors who value association with an iconic brand but can tolerate race-to-race unpredictability.

9. **Reliability has improved grid-wide since 2019, making F1 a more reliable advertising medium.** Average DNFs across the grid have trended downward since the 2019 improvement peak, with 2024 seeing McLaren achieve zero DNFs. Sponsors can now expect more guaranteed exposure per race weekend than at any point in the Hybrid Era.
10. **The undervalued gems cohort (Aston Martin, Haas, Williams, Alfa Romeo, Ferrari) represents a structured budget opportunity.** Five teams exhibit positive growth momentum with below-average investment scores. For CMOs with limited budgets seeking long-term growth plays, these teams offer the highest potential appreciation of sponsorship value.

11 Recommendations

Table 13: Recommendation framework: Insight → Action → Impact

#	Driving Insight	Recommendation	Expected Impact
R1	McLaren is Gold-tier with +229.5 momentum slope and zero 2024 DNFs	Initiate a multi-year primary sponsorship agreement with McLaren now, before their 2025 championship defence elevates market pricing	Estimated 2–3× brand exposure per dollar vs. current Red Bull rates; lock in below-market pricing before championship confirmation
R2	Diversified portfolio outperforms single-team concentration on risk-adjusted basis	Build a portfolio: McLaren (premium), Sauber (stable hedge), Aston Martin (growth play). Distribute budget 60:25:15 across the three tiers	Reduces brand exposure risk to ≈35% (vs. 100% with single-team), while maintaining premium association through McLaren
R3	Red Bull’s momentum slope is –93.5 with statistically confirmed decline	Re-negotiate any existing Red Bull agreements to include performance-based fee adjustments; redirect new budget away from Red Bull unless momentum reverses for 2+ consecutive seasons	Avoid overpaying by 20–30% for exposure that is declining relative to the 2022–2023 peak
R4	5 “Undervalued Gems” have positive momentum at below-average cost	Explore early-stage commercial agreements with Aston Martin or Haas at current rates before their 2025–2026 performance improvements are priced into the market	Potential 40–60% cost advantage vs. waiting for market recognition of team improvements
R5	Operational metrics (pit stop speed, DNF rate) are statistically validated investment signals	Require quarterly operational performance reports from sponsored teams as part of commercial agreements; include reliability KPI thresholds as contract performance clauses	Early warning system for team quality decline; provides contractual basis for rate renegotiation if reliability drops below agreed thresholds

12 Impact Estimation

The following estimates are based on reasonable assumptions from the analysis. Exact dollar values would require proprietary sponsorship contract data not publicly available.

12.1 R1: McLaren Premium Partnership

- McLaren achieved 609 points in 2024, compared to Red Bull at 537. In terms of podiums, McLaren had 21 vs. Red Bull's 18 in 2024.
- Each podium represents approximately 90 minutes of premium broadcast time with podium ceremony branding. McLaren delivered 3 extra podiums (approximately 4.5 additional hours of podium brand exposure).
- Based on media valuation frameworks standard in motorsport (typically \$5,000–\$15,000 per broadcast second for primary livery), the McLaren premium over Red Bull equates to an estimated additional \$1.5–4.5M in earned media value over a full season.

12.2 R2: Diversified Portfolio

- A portfolio spreading budget 60:25:15 across McLaren/Sauber/Aston Martin reduces single-team exposure risk from 100% to approximately 33% on any individual performance failure.
- Based on the Efficient Frontier analysis, this portfolio combination sits above the average return line while remaining below average volatility — the theoretical “efficient” allocation.

12.3 R3: Red Bull Renegotiation

- Red Bull's 2024 points (537) represent a –32% decline from 2023 (790). A sponsor paying 2023 rates for 2024 delivery received approximately 32% less exposure value than contracted.
- Performance-based contract clauses tied to finishing positions could have automatically adjusted fees by an estimated 15–25%, saving a mid-size sponsor \$500K–\$2M on a typical Formula 1 arrangement.

13 Limitations

1. **No financial data:** The Investment Scorecard is built entirely on performance metrics as proxies for brand value. It does not incorporate actual sponsorship fee data, which is proprietary and unavailable. The framework ranks relative investment quality, not absolute ROI.
2. **Team continuity gaps:** Teams that rebranded mid-era (Force India → Racing Point → Aston Martin; Toro Rosso → AlphaTauri → RB F1) are treated as separate entities. This understates the performance continuity of Red Bull's junior team programme and Aston Martin's lineage.
3. **COVID-19 anomaly (2020):** The 2020 season was 30% shorter than a standard season. YoY growth calculations around this year may be distorted (2020 → 2021 growth is inflated because 2020 was a shortened baseline).
4. **Driver vs. constructor conflation:** A significant portion of a team's performance variance is attributable to individual driver talent rather than team quality (e.g., the Max Verstappen effect at Red Bull). Our constructor-level analysis cannot separate driver contribution from team infrastructure contribution.

5. **Recency of data:** The dataset extends to the end of 2024. Significant developments in 2025 (new regulations, driver transfers, team technical changes) are not captured.
6. **Pit stop sample sizes:** Some teams have limited pit stop records in specific seasons (particularly smaller teams with earlier exits). Low sample sizes can produce misleading average duration values for those specific data points.

14 Future Scope

1. **Predictive modelling:** A time-series forecasting model (e.g., ARIMA or Prophet) applied to each constructor's points trajectory would enable the dashboard to display a predicted 2025 championship position, giving sponsors a forward-looking ranking rather than a backward-looking scorecard.
2. **Driver-adjusted team ratings:** By computing a “driver premium” (how much better each driver performs vs. the team's baseline), it would be possible to separate team infrastructure quality from individual talent. This would produce a purer measure of team investment quality that is less sensitive to driver transfer news.
3. **Real-time data integration:** The Ergast API provides end-of-season historical data. Integrating with the official F1 API or a live timing provider would enable a real-time dashboard that updates after every race, providing sponsors with in-season monitoring of their investment.
4. **Sponsor ROI linkage:** If brand awareness survey data or social media sentiment data were linked to race results, it would be possible to build a true ROI model connecting on-track performance to off-track brand metric changes.
5. **Circuit-level analysis:** Some circuits (Monaco, Singapore, Monza) receive disproportionately high global media attention. A circuit-weighted scoring model would distinguish between points scored at high-viewership events vs. lower-viewership rounds.
6. **New data sources:** Social media follower counts, team merchandise revenue, and championship broadcast viewership ratings would significantly enrich the investment model beyond pure race performance metrics.

15 Conclusion

Formula 1 is a \$2.6 billion sponsorship market where investment decisions are too often driven by historical prestige rather than forward-looking evidence. This project addressed that gap by building a complete analytical pipeline — from raw historical race data to an interactive, decision-ready Investment Scorecard — using 4,626 race records spanning 11 seasons, 20 constructors, and 32 circuits across the Modern Hybrid Era.

The analysis reveals that **McLaren represents the highest-value sponsorship opportunity on the grid**, with statistically confirmed positive momentum, perfect 2024 reliability, and a composite Investment Score of 63.02 — the only Gold-tier team.

Conversely, Red Bull’s momentum slope of –93.5 points/year and a –32% season-on-season decline confirm that its premium pricing is based on a past dominance cycle that has already ended. Five “undervalued gems” (Aston Martin, Haas, Williams, Alfa Romeo, Ferrari) offer growing performance at below-average investment costs, representing the most compelling budget-conscious opportunities.

The highest-impact recommendation is to build a diversified F1 sponsorship portfolio — mirroring financial portfolio construction — with McLaren as the growth anchor, Sauber as the risk hedge, and one undervalued gem as the high-upside growth play. This structure maximises brand exposure across the grid while dramatically reducing the performance risk of any single-team concentration.

The methodology developed in this project — a five-dimension KPI framework combined with statistically validated trend analysis — provides a replicable, data-driven alternative to intuition-based sponsorship decisions, applicable to any future season as the sport continues to evolve.

Appendix

This appendix gathers the supporting reference material used in the analysis. It is structured for quick lookup, so readers can verify definitions, outputs, tools, and contribution details without interrupting the main argument of the report.

A Data Dictionary (Full Column Definitions)

Column	Source Table	Type	Description
resultId	results	int	Primary key for each driver-race result
raceId	results	int	FK to races table
driverId	results	int	FK to drivers table
constructorId	results	int	FK to constructors table (main grouping key)
grid	results	int	Starting grid position (0 = pit lane start)
positionOrder	results	int	Final classification order (always filled)
points	results	float	Championship points scored
laps	results	int	Number of laps completed
statusId	results	int	FK to status table
year	races	int	Season year (filtered: ≥ 2014)
round	races	int	Race number within season
race_name	races	str	Full Grand Prix name
race_date	races	date	Race date (YYYY-MM-DD, converted to datetime)
constructor_name	constructors	str	Official team name (e.g., McLaren, Ferrari)
constructor_nationality	constructors	str	Team registered nationality
forename	drivers	str	Driver first name

Column	Source Table	Type	Description
surname	drivers	str	Driver last name
driver_nationality	drivers	str	Driver nationality
status	status	str	Finish status text (e.g., Finished, Engine)
circuit_name	circuits	str	Official circuit name
circuit_location	circuits	str	City of circuit
circuit_country	circuits	str	Country of circuit
lat / lng	circuits	float	GPS coordinates for Tableau map
position_delta	Engineered	int	grid – positionOrder (race-day gain/loss)
finished	Engineered	int	1 = classified finisher, 0 = DNF
driver_name	Engineered	str	forename + surname

B Tableau-Ready Output Files

File	Rows	Cols	Used In
constructor_season_summary.csv	112	21	Dashboards 1, 2, 3
race_detail.csv	4,626	32	Drill-down tables
pit_stop_summary.csv	112	9	Dashboard 3
investment_scorecard.csv	12	17	Dashboards 1, 2

C Python Library Stack

Library	Version	Purpose
pandas	2.3.0	Data manipulation and aggregation
numpy	2.3.0	Numerical operations
matplotlib	3.10.3	Base plotting engine
seaborn	0.13.2	Statistical chart aesthetics
scipy.stats	1.x	t-Tests, linregress, Levene's test
sklearn	1.x	LinearRegression, StandardScaler

D Points Concentration Data Table (EDA Output)

Table 15: Full season-by-season points concentration (from 03_eda.ipynb)

Year	Top Team	Total Pts	Top Pts	Top 1%	Top 2 Pts	Top 2%
2014	Mercedes	2,020	701	34.70%	1,106	54.75%
2015	Mercedes	1,919	703	36.63%	1,131	58.94%
2016	Mercedes	2,121	765	36.07%	1,233	58.13%
2017	Mercedes	2,020	668	33.07%	1,189	58.91%
2018	Mercedes	2,062	655	31.77%	1,226	59.46%
2019	Mercedes	2,140	739	34.53%	1,243	58.08%
2020	Mercedes	1,719	573	33.33%	892	51.89%
2021	Mercedes	2,208	613	27.79%	1,199	54.31%
2022	Red Bull	2,350	759	32.30%	1,313	55.87%
2023	Red Bull	2,458	860	34.99%	1,269	51.63%
2024	McLaren	2,659	666	25.05%	1,318	49.57%

Contribution Matrix

This section clarifies team ownership across the project so evaluators can quickly see who led each major deliverable.

Core = major contribution **Assist** = supporting contribution

Table 16: Team Contribution Matrix

Member	Dataset	ETL	EDA	Stats	Tableau	Report	PPT
Priyanshu Tomar	Core	Core	Assist	Assist	Assist	Core	Core
Piyush	Assist	Assist	Assist	Core	Assist	Assist	Core
Kushagra Maheshwari	Assist	Assist	Core	Assist	Core	Assist	Assist
Thejas	Assist	Core	Assist	Core	Assist	Assist	Assist
Anurag Kumar	Core	Assist	Assist	Assist	Assist	Core	Assist
Aditya Srivastava	Assist	Assist	Assist	Assist	Core	Assist	Core

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts committed to https://github.com/r0c0y/SectionC_G13_F1.

Team Lead: Priyanshu Tomar **Date:** April 29, 2026

Newton School of Technology | Data Visualization & Analytics | Capstone 2
Section C | Group G13 | April 2026
